

### General Information

A revised cylinder head casting offering more efficient air flow, a durability improvement, and standardization to one head casting was introduced in May 1991, Engine Serial Number, first 33119713. This casting can be identified by the cast-in part number on the top of the head. The revised casting can be used in place of all previous KV cylinder head castings.

#### Premium Number 1

- This cylinder head assembly, Part Number 3811985, is used on high load factor and constant speed applications, such as marine, base load or prime power generator drives, some drilling machines, and hydraulic excavators. This cylinder head contains powdered metal valve guides, valve rotators, shorter springs, wear resistant Tribaloy™ intake valves and valve inserts, and high temperature Inconel™ exhaust valves.

#### Premium Number 2

- This cylinder head assembly, Part Number 3811986, is used to update engines that do **not** require the Tribaloy™ intake valve. This cylinder head is used on all K series engines with a design rating of 101 horsepower (HP) per cylinder or more, that do **not** require the Premium Number 1 cylinder head assembly.

#### Improved Standard Cylinder Head

- The improved standard cylinder head assembly, Part Number 3811987, is used on all other K series engines with a maximum design rating of 100 horsepower (HP) per cylinder or less, but **not** for use on standby generator sets. This cylinder head contains standard intake valves, Stellite™ 6 exhaust valves, standard valve inserts, valve rotators, and springs.

#### Standard Cylinder Head

- This cylinder head assembly, Part Number 3811988, is used in standby generator sets, **not** prime power or base load units. This assembly contains standard intake valves, Stellite™ 6 exhaust valves, and standard valve inserts, but does **not** contain valve rotators.

#### Special Application Cylinder Heads

- This cylinder head assembly, Part Number 3811989, is for use on locomotive applications and is used on some hydraulic excavators. It contains standard valves and flat-top cast iron valve guides with seals on all four valve guides, but does **not** contain rotators. These valve seals are used in applications where there is a large percentage of idle time that can cause fuel slobber.

**Do not downgrade cylinder head assemblies such as replacing Premium Number 1 cylinder head assembly, Part Number 3081070, with Premium Number 2 cylinder head assembly, Part Number 3081071. Upgrading of cylinder head assemblies is allowable.**

After thousands of hours of severe use, small cracks can begin to appear on the combustion surface between the valves and around the injector bore. Cylinder heads with cracks that extend into, but **not** across, the valve seat insert bore of the head can be used again.

**Note : Cummins Inc. does not recommend that cracks be repaired by pinning or welding**

Because of the large surface area of the valves, the valve depth in the head is critical to the operation of the engine. Valves installed to a depth greater than 0.05 mm [ 0.002 in ] will result in excessive white smoke when the engine is operating in cold temperatures.

The cylinder head castings, Part Numbers 3009088, 3176619, 3025042, and 3081064, are acceptable for rebuild and reuse.

When replacing cylinder head castings, Part Numbers 3009088 or 3176619, with casting, Part Number 3081064, on engines built to Control Parts List (CPL) -0801, add Field Fix Number 183 to the dataplate.

Cylinder head castings, Part Numbers 205010, 205015, and 205016, **must** have the injector bore machined to use Premium K injectors.

**⚠ CAUTION ⚠**

**Do not machine the injector bores with the cylinder heads installed on the engine. Machining chips can fall into the power cylinder and cause damage to the piston rings and cylinder liner. Any of these castings remanufactured by ReCon® after August 1993 will have the injector bores machined again.**

Specific generator drive applications require cylinder head casting, Part Number 3050491, Part Number 3068171, or Part Number 3081064 (see to Table I). Use cylinder head castings, Part Number 3050491 and Part Number 3068171, **only** on those CPL Numbers listed in Table II. These castings are **not** to be rebuilt or reused unless on the same engine and in the same application from which it was removed. These cylinder head castings can **not** be purchased through ReCon®. If any of the above mentioned CPL Numbers require a rebuilt cylinder head, one of the following assembly numbers, which include cylinder head casting, Part Number 3081064, **must** be purchased new.

Cylinder head assembly, Part Number 3811985, for prime power and base load generator drives.

Cylinder head assembly, Part Number 3811988, for standby power generator drives **only**.

**Table I - Cylinder Head Casting History**

| Old Part Number | Date Used | Replaced by Part Number |
|-----------------|-----------|-------------------------|
| 205010          | 1972-1976 | 205016                  |

**Table I - Cylinder Head Casting History**

| Old Part Number | Date Used    | Replaced by Part Number |
|-----------------|--------------|-------------------------|
| 205015          | 1976         | 3007207                 |
| 205016          | 1976         | 3007207                 |
| 3007207         | 1976-1977    | 3009088                 |
| 3009088         | 1977-1985    | 3176619                 |
| 3176619         | 1985-1993    | 3081064                 |
| 3050491         | 1985-1993    | 3081064                 |
| 3068171         | 1992-1993    | 3081064                 |
| 3081064         | 1993-Present | N/A                     |

**Table II - Cylinder Head Castings, Part Numbers 3050491 and 3068171, Engine Models**

| CPL Number | Engine Model | Option Number |
|------------|--------------|---------------|
| CPL-1251   | KTA38-G3     | PP6123        |
| CPL-1541   | KTA38-G3     | PP6188        |
| CPL-1495   | KTA38-G4     | PP1495        |
| CPL-1542   | KTA38-G4     | PP6189        |
| CPL-1497   | KTA38-G5     | PP1497        |
| CPL-1543   | KTA38-G5     | PP6190        |
| CPL-1219   | KTA50-G3/G4  | PP6121        |
| CPL-1528   | KTA50-G3/G4  | PP6191        |
| CPL-1254   | KTTA50-G2    | PP6124        |
| CPL-1544   | KTTA50-G2    | PP6192        |

**Table III - Cylinder Head Assemblies Supersession**

| Head Casting Part Number<br>3081064 Assembly Numbers | Cylinder Head Casting<br>Assembly Type | Superseded Assembly Part<br>Number |
|------------------------------------------------------|----------------------------------------|------------------------------------|
| 3811985                                              | PREMIUM Number 1                       | 3072438                            |
|                                                      |                                        | 3627612                            |
|                                                      |                                        | 3627752                            |
|                                                      |                                        | 3081070                            |
| 3811986                                              | PREMIUM Number 2                       | 3074622                            |
|                                                      |                                        | 3627753                            |
|                                                      |                                        | 3081071                            |
| 3811987                                              | IMPROVED STANDARD                      | 3074623                            |
|                                                      |                                        | 3081072                            |

**Table III - Cylinder Head Assemblies Supersession**

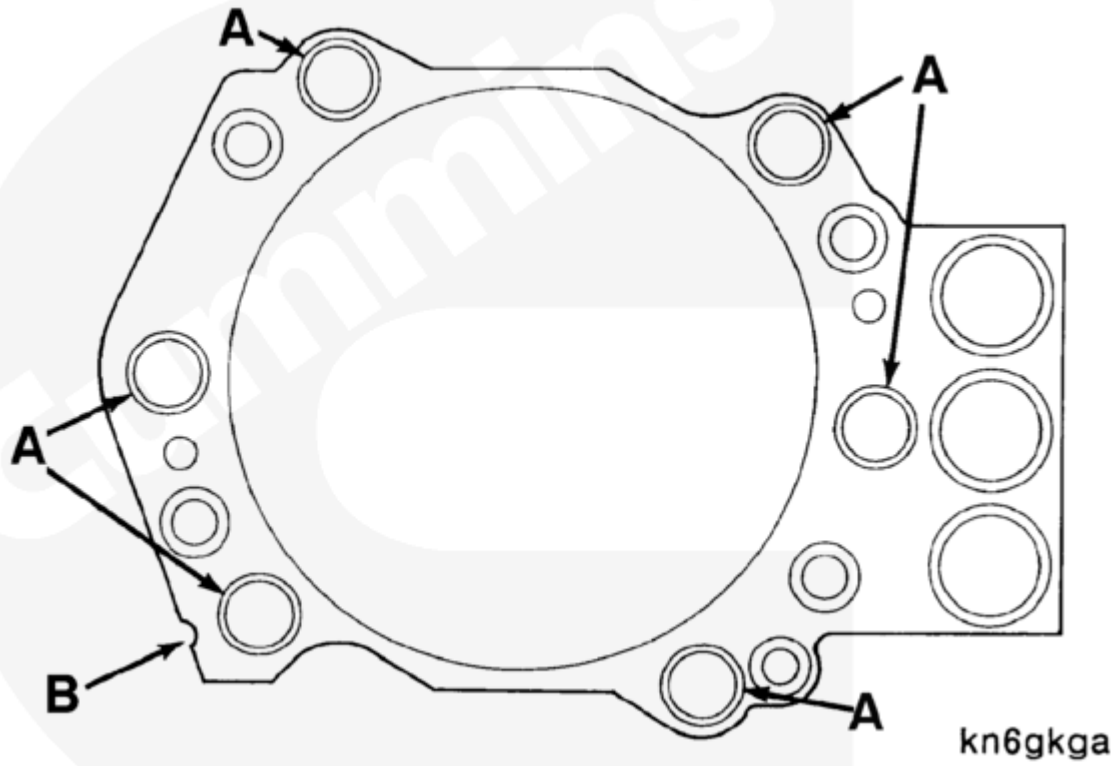
| <b>Head Casting Part Number<br/>3081064 Assembly Numbers</b> | <b>Cylinder Head Casting<br/>Assembly Type</b> | <b>Superseded Assembly Part<br/>Number</b> |
|--------------------------------------------------------------|------------------------------------------------|--------------------------------------------|
| 3811988                                                      | STANDARD                                       | 3068402                                    |
|                                                              |                                                | 3068404                                    |
|                                                              |                                                | 3080113                                    |
|                                                              |                                                | 3081073                                    |
|                                                              |                                                | 3177529                                    |
|                                                              |                                                | 3081074                                    |
|                                                              |                                                | 3628226                                    |
|                                                              |                                                | 3081075                                    |
|                                                              |                                                | 3811990                                    |

Cylinder head castings, Part Number 3050491, 3068171, and 3081064, can be identified by the cast-in part number and divider wall in the exhaust port. The divider wall separates the gases from both exhaust valves.

A revised cylinder head gasket, Part Number 3628407, was used on all K2000E, K1800E, and construction engines built since December 1992, Engine Serial Number (ESN) first 33123381. This revised standard thickness gasket is also to be used for all K38 and K50 service requirements.

The gasket has grommets (A) surrounding all of the cylinder head capscrew holes. The grommets prevent dirt, water, and coolant from reaching the cylinder head capscrews, extending capscrew life and reusability.

A notch (B) in the gasket steel plate can be used to identify the grommited gasket after installation on the engine.



Also, a revised 0.508 mm [ 0.020 in ] oversize gasket with grommets around all cylinder head capscrew holes has been released for service usage when material has been machined from the cylinder block top deck.

Anti-swirl plates, also called air deflectors or swirl plates, are used in some cylinder head assemblies. The plates are installed under the intake valve inserts.

Anti-swirl plates are used to increase performance by mixing the air and fuel.

When the K38 engine was introduced, the cylinder head did **not** contain an anti-swirl plate.

Two anti-swirl plates have been used. These anti-swirl plates are **not** interchangeable. Use anti-swirl plate, Part Number 3009109, with cylinder head casting, Part Numbers 3007207, 3009088, or 3176619. Use anti-swirl plate, Part Number 206363, with any of the other K engine cylinder head castings that require an anti-swirl plate.

For optimum performance of the engine, always use the correct anti-swirl plate or the insert spacer plate. See CPL, Bulletin Number 3379133, for the correct cylinder head casting number for the engine being serviced.

**Note : In order to meet EPA certifications, certain castings must use anti-swirl plates.**

ReCon® has implemented a new stamping code for all K and QSK cylinder heads.

The stamp is a diamond shape located between the short port exhaust valve and long port intake valve on the rocker lever side of the cylinder head. The marking, or lack of a marking, inside the diamond shape indicates the size of the injector seal.

- An empty diamond shape indicates that the ReCon® cylinder head is equipped with a standard size injector seal.
- One mark inside the diamond shape indicates the cylinder head is equipped with a 0.010 oversize injector seal.
- Two marks inside the diamond shape indicate that the cylinder head is equipped with a 0.020 oversize injector seal.
- Three marks inside the diamond shape indicate that the cylinder head is equipped with a 0.030 oversize injector seal.

For K cylinder heads to achieve the injector protrusion within specifications, an oversize injector seal can be used. Reference the table below.

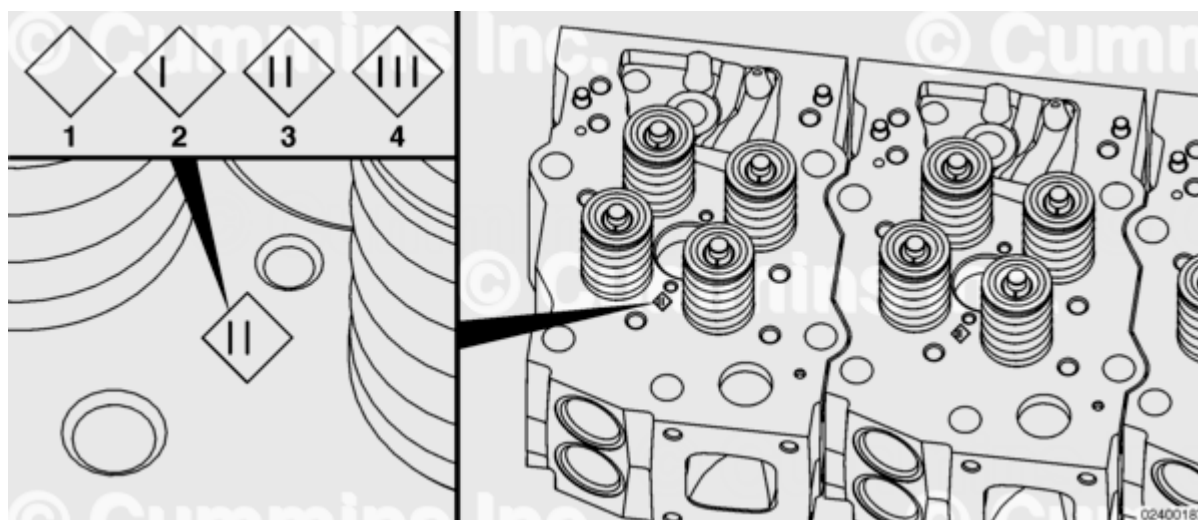
#### K Cylinder Head Seal Part Numbers

| Injector Seal Part Number | Injector Seal Size |
|---------------------------|--------------------|
| 207244                    | Standard           |
| 3001658                   | 0.010 oversize     |
| 3001659                   | 0.020 oversize     |
| 3001660                   | 0.030 oversize     |

In K engines with mechanically actuated injectors, an appropriate injector seal is used.

In QSK engines, **only** MCRS injectors are used.

**Note :** No injector protrusion checks or protrusion specifications on electronically actuated injectors are required, as injector protrusion can not be adjusted.



Injector Seal Markings Location

#### Injector Seal Markings

1. Standard
2. 0.010 oversize
3. 0.020 oversize

## Preparatory Steps

### with Mechanically Actuated Injector

#### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

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ck800wa

#### **⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [ 120°F ] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**

#### **⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

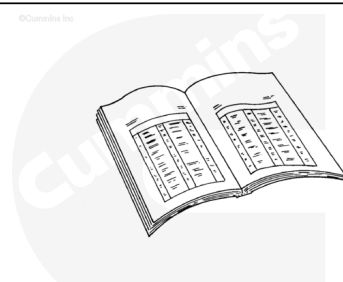
- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Disconnect the air supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html>)

- Remove the aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-002.html>)
- Remove the fuel manifolds. Refer to Procedure 006-022 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-022.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html>)
- Remove the valve push rods, injector push rod, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html>)
- Remove the injector. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html>)

## with Electronically Actuated Injector

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



ck800wa

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### **⚠ WARNING ⚠**



**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

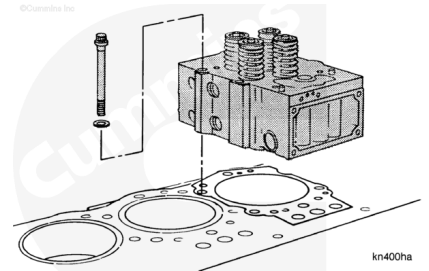
- Disconnect the batteries. Refer to the OEM service manual.
- Disconnect the air supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html>)
- Remove the high pressure fuel lines to the injectors. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html>)
- Remove the fuel drain lines. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-013.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html>)
- Remove the valve push rods, and crossheads. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html>)
- Remove the injector. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html>)

## Remove

### with Mechanically Actuated Injector

#### **⚠ CAUTION ⚠**

The cylinder head capscrews and washers are special parts. Do not damage the parts by throwing them against each other or in a container. Damage to the parts can result in engine failure.

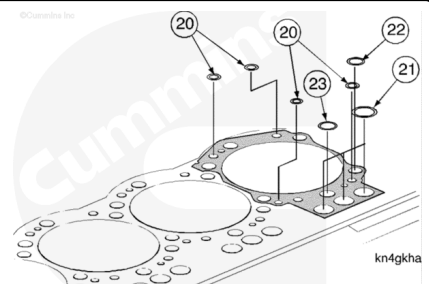


**Note :** There are six capscrews for each cylinder head.

Remove all of the cylinder head capscrews and washers.

#### **⚠ WARNING ⚠**

This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



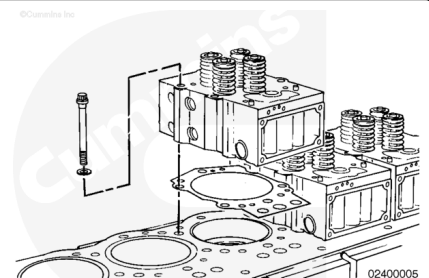
Remove all the cylinder heads.

Remove all the cylinder head gaskets, seals, and grommets.

### with Electronically Actuated Injector

#### **⚠ WARNING ⚠**

This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the six cylinder head capscrews.

Remove the cylinder head and gasket.

Record the cylinder head gasket part number to determine if the gasket is a standard or oversized thickness.

Discard the gasket.

# Clean

## with Mechanically Actuated Injector

### **⚠ WARNING ⚠**

**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**

### **⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

### **⚠ WARNING ⚠**

**When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



Steam clean the cylinder head.

Blow out all capscrew holes, fuel passages, and oil passages with compressed air.

Remove the gasket material from the rocker lever housing surface.

Use a gasket scraper and solvent to remove the heavy dirt and debris from the cylinder head gasket surface and the exhaust manifold gasket surface.

## with Electronically Actuated Injector

### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



Clean the cylinder heads with solvent, Part Number 3824421, or equivalent.

Check for broken springs or other damage.

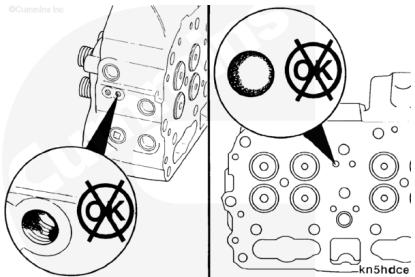
## Inspect for Reuse

### with Mechanically Actuated Injector

**Note :** The inspection specifications for cylinder heads used on natural gas fueled engines is the same as for the mechanically actuated injector except where noted.

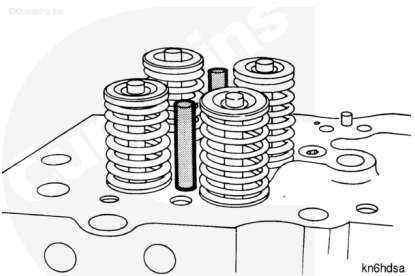
Inspect the fuel drillings and the water passages for restrictions or foreign material.

Remove any obstructions.



**Note :** If cracked or damaged parts are found, the cylinder head must be rebuilt.

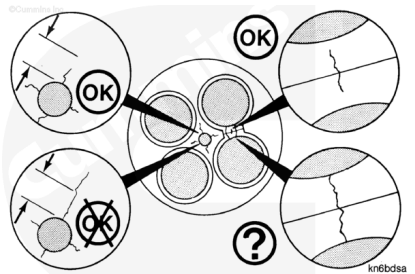
Inspect the valves and the valve springs for cracks, bent or broken valve stems, broken valve springs, or other damage.



Check for cracks on the combustion surface.

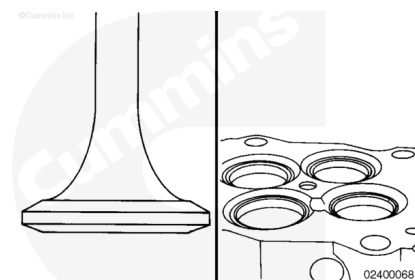
**Note :** If a crack around the injector bore exceeds the maximum length, the cylinder head must be replaced.

| Maximum Allowable Crack Length |     |      |
|--------------------------------|-----|------|
| mm                             |     | in   |
| 6                              | MAX | 0.25 |



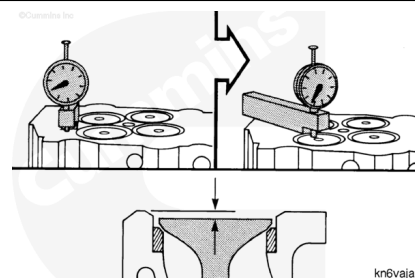
**Note :** Both ends of the crack between the valve must be visible. If one end of the crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder is questionable. To be sure that the cylinder head is reusable, remove the valve and valve seat.

Inspect the valves for indications of leakage or burning. If indications of leakage or burning are found, the valves and the seats **must** be replaced or resurfaced.



Use a depth gauge, Part Number 3164438, or equivalent. Measure the valve head depth. The valve head protrusion from the combustion face is **not** acceptable.

If the valve depth exceeds specifications, the cylinder head **must** be rebuilt.

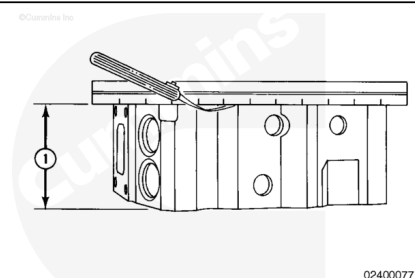


#### Maximum Allowable Valve Depth from the Combustion Face

| mm   |     | in    |
|------|-----|-------|
| 0.00 | MIN | 0.00  |
| 0.51 | MAX | 0.020 |

Check the flatness of the cylinder head with a straightedge and feeler gauge.

The cylinder head **must** be resurfaced or replaced if a feeler gauge larger than 0.08 mm [ 0.003 in ] will fit between the straightedge and the cylinder head. The cylinder head can be resurfaced as long as the head thickness measurement is within specifications.



#### Cylinder Head Thickness

|                                 | mm     |     | in    |
|---------------------------------|--------|-----|-------|
| Service Limit (1)               | 119.76 | MIN | 4.715 |
| New Cylinder Head Thickness (1) | 120.52 | MIN | 4.745 |
|                                 | 120.78 | MAX | 4.755 |

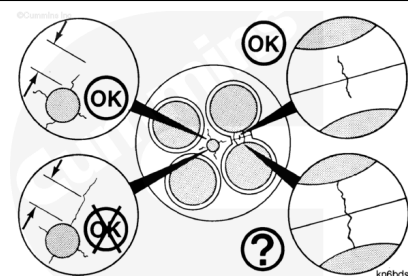
If the cylinder head is resurfaced, make sure the injector protrusion and the valve depth in the head are properly adjusted.

**Note :** The cylinder head can be resurfaced **only** on mechanically actuated injectors, if the head thickness measurement is within specification.

## with Electronically Actuated Injector

Check for cracks on the combustion surface.

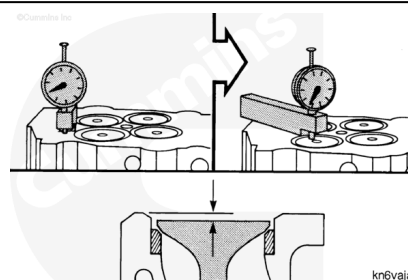
If a crack around the injector bore exceeds the maximum length, the cylinder head **must** be replaced.



| Maximum Allowable Crack Length |     |      |
|--------------------------------|-----|------|
| mm                             |     | in   |
| 6                              | MAX | 0.25 |

Both ends of a crack between the valves **must** be visible. If one end of a crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder is questionable. To be sure that the cylinder head is reusable, remove the valve and valve seat.

Measure the valve head depth in the cylinder head with depth gauge, Part Number 3164438, or equivalent.

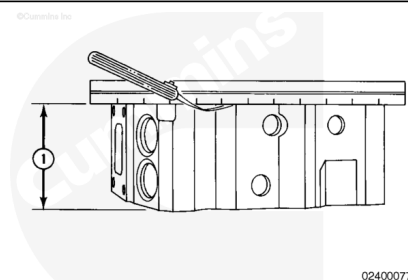


| Maximum Allowable Valve Depth from the Combustion Face |     |       |
|--------------------------------------------------------|-----|-------|
| mm                                                     |     | in    |
| 0.00                                                   | MIN | 0.000 |
| 0.51                                                   | MAX | 0.020 |

If the valve head depth is **not** within specifications, the cylinder head **must** be reconditioned.

Check the flatness of the cylinder head with a straightedge and feeler gauge.

The cylinder head **must** be replaced if a feeler gauge larger than 0.08 mm [ 0.003 in ] will fit between the straightedge and the cylinder head.



| Cylinder Head Thickness |    |  |    |
|-------------------------|----|--|----|
|                         | mm |  | in |
|                         |    |  |    |

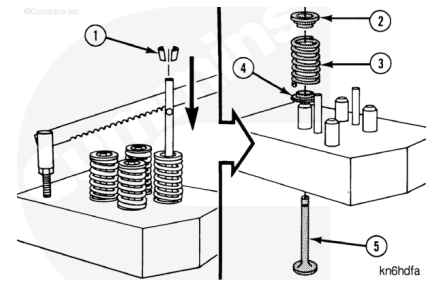
|                                             |        |     |       |
|---------------------------------------------|--------|-----|-------|
| New<br>Cylinder<br>Head<br>Thickness<br>(1) | 120.52 | MIN | 4.745 |
|                                             | 120.78 | MAX | 4.755 |

**Note :** The cylinder heads with Electronically Actuated Injectors can not be resurfaced, as injector protrusion can not be adjusted.

## Disassemble

### **⚠ WARNING ⚠**

**The valve springs are under compression. Use care when using the valve spring compressor. Personal injury can result if the tool slips from your hands.**



Use one of the valve spring compressors listed below to remove the valve springs.

- Valve spring compressor, Part Number 3163606, or equivalent
- Valve spring compressor stand, Part Number ST-1022, or equivalent
- Valve spring compressor plate, Part Number 3163177, or equivalent, can be used with valve spring compressor, Part Number 3163606, or equivalent, and valve spring compressor stand, Part Number ST-1022, or equivalent, to remove four springs at the same time
- Air operated valve spring compressor, Part Number 3375960.
- Spark plug adapter replacer, Part Number 3164898

The intake and exhaust valves are different.

Mark the valves for location prior to removal to aid in assembly.

Remove the listed parts:

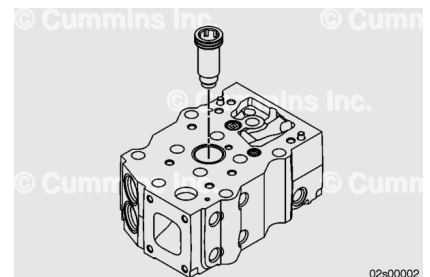
1. Valve collet
2. Valve spring retainer

3. Valve spring
4. Valve spring rotator/or guide
5. Valve
6. Valve stem seal (**not** shown).

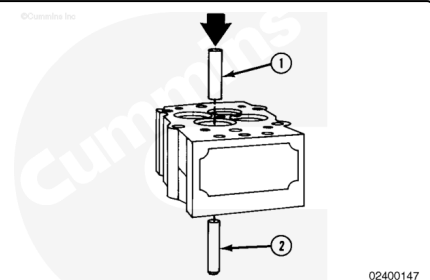
Remove the spark plug adapter and gasket or washer.

Refer to Procedure 002-015 in Section 2

(/qs3/pubsys2/xml/en/procedures/445/445-002-015-tr.html)  
in Bulletin 5504185 GTA38E G CMEICS G114 and GTA50E G  
CMEICS G115 Service Manual



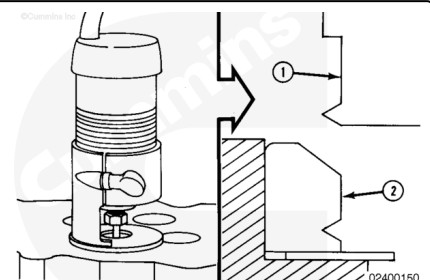
If the valve guide is replaced, the valve seat insert for the valve guide **must** be measured for run out. It is likely the seat **must** also be machined or replaced.



**Note : Only replace the valve seat if it is not within specification.**

If necessary, a groove can be machined in the valve seat insert (2) to allow the valve seat extractor (1) to be used.

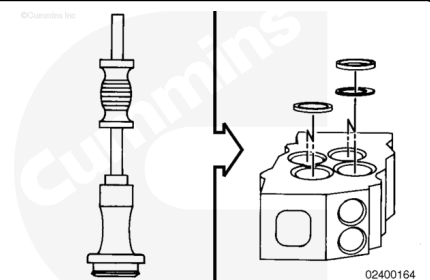
Machine a groove into the valve seat insert, as close to the bottom of the bore as possible, with valve seat grooving kit, Part Number 3376405, or equivalent.



Remove the valve seat with the appropriate valve seat extractor listed below and slide hammer, Part Number 3376799:

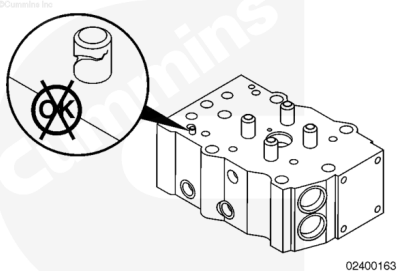
- Exhaust seat extractor, Part Number ST-1323-1
- Intake seat extractor, Part Number 3376799.

Remove and discard the anti-swirl plate, if used.



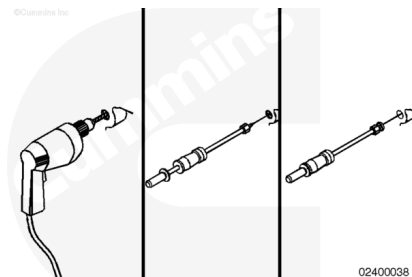
Remove the dowel pins **only** if they are damaged or **not** the correct height.





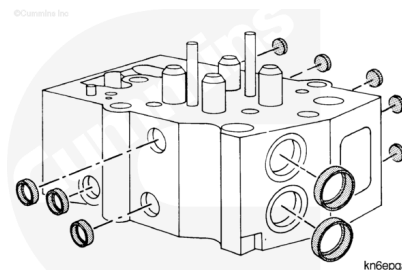
**Note : Do not remove the cup plugs if a pressure test is to be performed.**

Use a drill, sheet metal screw, and a slide hammer from the light duty puller kit, Part Number 3375784, or equivalent, to remove the cup plugs.



Cup plugs **must** be removed from the head casting for cleaning purposes.

Remove and discard the 10 cup plugs.

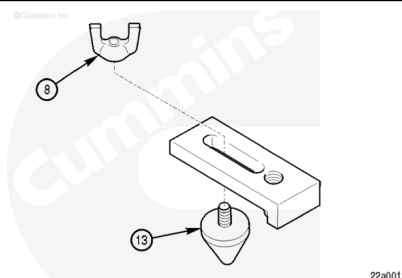


## Pressure Test

### Preferred Method

Use the cylinder head leak test kit, Part Number 3164341, or equivalent, with the pressure regulator valve kit, Part Number 3164231, or equivalent, to pressure test the cylinder head.

Assemble the wing nut (8) to the clamp and plug (13).

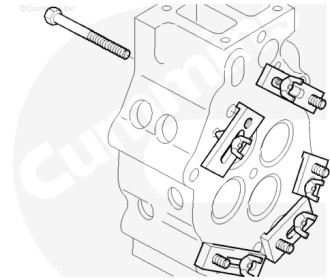


Install the clamp and plug assembly into the cylinder head to block the coolant passages.

Insert the appropriate capscrew through the valve guide side of the cylinder head and thread the capscrew into the clamp.

Tighten the capscrew.

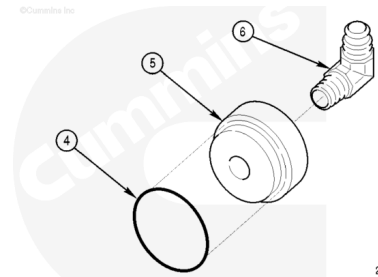
**Torque Value:** 20 n•m [ 177 in-lb ]



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Install the o-ring (4) into the o-ring groove of the test adapter (5).

Thread the test adapter elbow (6) into the test adapter (5).



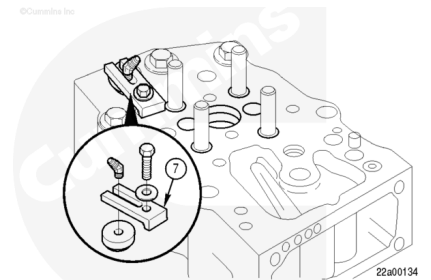
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Place the test adapter assembly into the cylinder head.

Install the clamp (7) over the test adapter assembly and secure it with the appropriate capscrew and washer.

Tighten the capscrew.

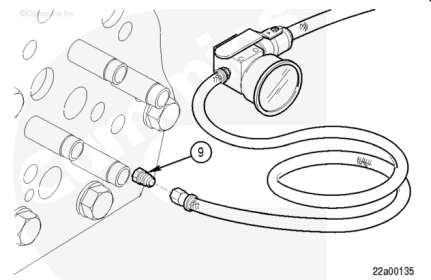
**Torque Value:** 54 n•m [ 40 ft-lb ]



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Connect the pressure regulator valve kit to the pipe nipple.

Connect the pressure regulator valve kit to shop air and adjust the air pressure.



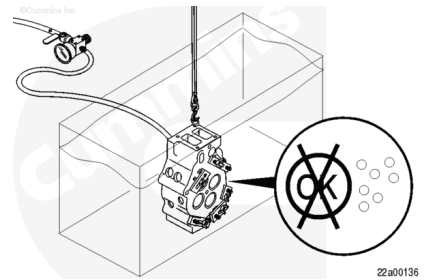
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**Measurements**

|                             | kpa | psi |
|-----------------------------|-----|-----|
| Pressure<br>Regulator Valve | 345 | 50  |

**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



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Attach an eye bolt to the cylinder head.

Use a suitable lifting device to lift the cylinder head.

Immerse the cylinder head into a tank of water.

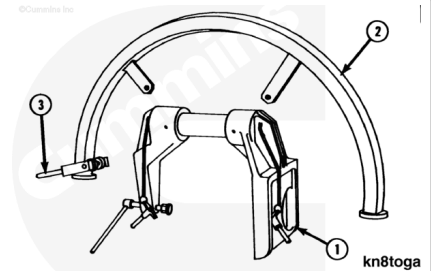
Check for bubbles on the lower deck. Carefully inspect the area of the spark plug hole.

If the cylinder head leaks, it **must** be repaired or replaced.

## Optional Method

One method to pressure test the cylinder head is to use the hydrostatic tester, Part Number ST-1012, or equivalent, with the water test adapter plate, Part Number 3375070.

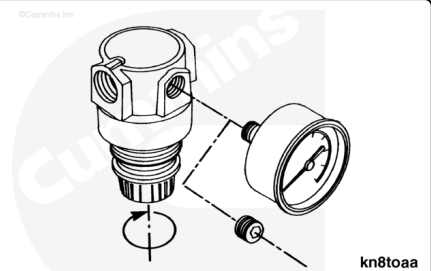
The following steps outline this method.



Remove one of the two plugs from the pressure regulator.

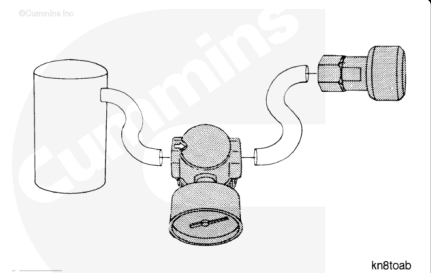
Install the pressure gauge in the regulator.

Turn the adjusting knob on the regulator **counterclockwise** as far as it will turn.



Install the pressure regulator between the air supply and the quick disconnect fitting.

The arrow on the top of the pressure regulator **must** point in the direction of the air flow (toward the quick disconnect fitting).

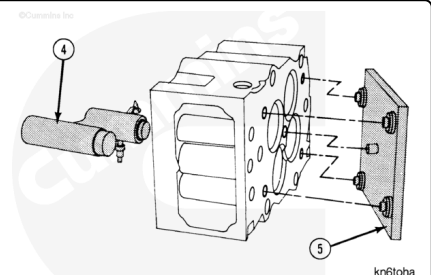


With the head positioned as illustrated, assemble the adapter plates.

The guide pins on the lower plate (5), Part Number 3375070, or equivalent, **must** fit into the water passages.

The o-ring on the upper plate (4) creates a seal on the upper water passage.

The other end of the plate fits into the injector bore.

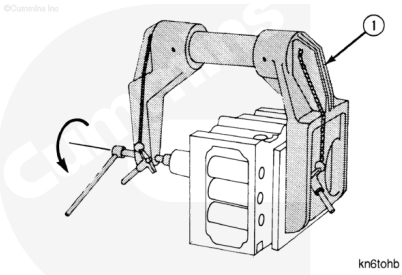


Place the clamping assembly (1) over the head and adapters.

The guide pins on the clamp **must** fit into the holes in the adapter plate.

Tighten the clamp on the cylinder head.

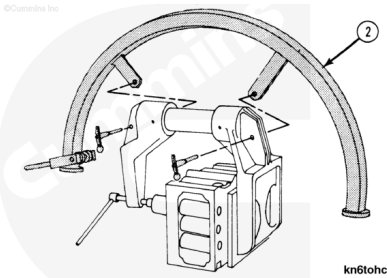
Connect the regulated air supply hose to the upper adapter fitting.



Attach the lifting arm (2) to the clamp assembly with the lock pins.

One mounting location on each piece is color-coded red.

Match the red locations.



**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

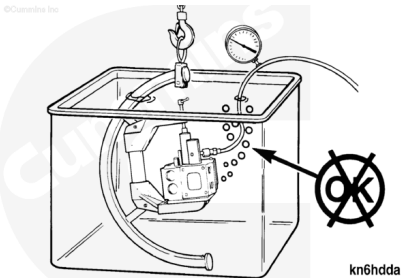
Adjust the air pressure.

| Air Pressure |     |     |
|--------------|-----|-----|
| kpa          |     | psi |
| 552          | MAX | 80  |

Lower the cylinder head into the tank of water.

Bubbles indicate an air leak.

If the cylinder head leaks, it **must** be repaired or replaced.

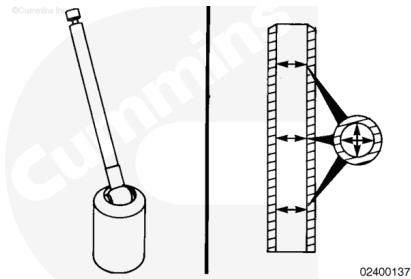


## Clean and Inspect for Reuse

**Note :** The specifications for natural gas heads are the same for diesel heads except where noted.

Check the valve guide for chips and cracks.

Measure the inside diameter of the valve guide with a ball gauge or a dial bore indicator.

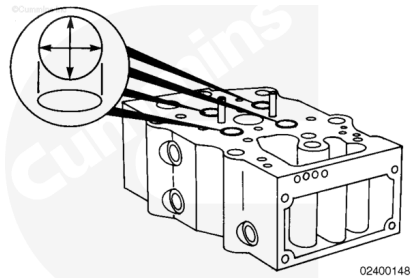


| Valve Guide Inside Diameter (Installed) |     |       |
|-----------------------------------------|-----|-------|
| mm                                      |     | in    |
| 12.600                                  | MIN | 0.496 |
| 12.667                                  | MAX | 0.499 |

The first 13 mm [ 0.50 in ] from the top or the bottom of the valve guide inside diameter can be too large and the valve guide still be reusable. If, at 13 mm [ 0.500 in ] or greater from the top or bottom, the inside diameter is out of tolerance, the valve guide **must** be replaced.

**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**



Clean the part with solvent, Part Number 3824421, or equivalent.

Measure the valve guide bore inside diameter.

| Valve Guide Bore Inside Diameter |     |       |
|----------------------------------|-----|-------|
| mm                               |     | in    |
| 21.425                           | MIN | 0.844 |
| 21.450                           | MAX | 0.845 |

If the valve guide bore is **not** within specifications, ream the bore for an oversize valve guide.

The oversize valve guides are oversize on the outside diameter **only**. The bore in the valve guide is **not** oversize.

Valve guides are available in two oversizes:

- 0.25 mm [ 0.010 in ]
- 0.38 mm [ 0.015 in ].

Ream the valve guide bore to maintain a press fit within 0.028 to 0.067 mm [ 0.001 to 0.003 in ] between the valve guide and the valve guide bore.

Clean the valve seat insert bore.

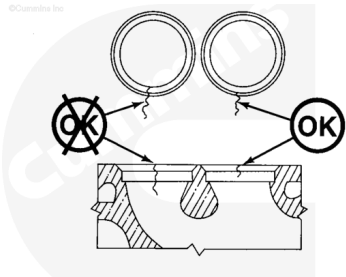
Check the length of any cracks extending into the valve insert bore.

The cylinder head **must** be replaced if a crack extends into the bottom of the bore.

At times it is possible to remove the crack by machining the cylinder head to use an oversize valve seat insert.

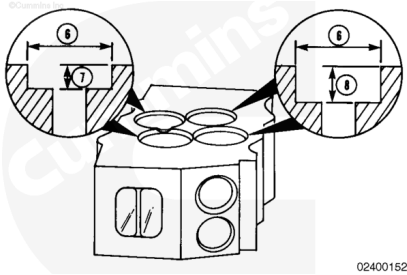
Use valve insert counterbore cutter kit, Part Number ST-257, or equivalent, along with the appropriate cutter listed below:

- ST-1310-1 (0.010 oversize)
- ST-1310-2 (0.020 oversize)
- ST-1310-3 (0.030 oversize)
- ST-1310-4 (0.040 oversize).



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Measure the inside diameter and the depth of the valve seat insert bore.



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Valve Seat Insert Bore Inside Diameter (6)

|                    | mm    |     | in    |
|--------------------|-------|-----|-------|
| Exhaust and Intake | 60.37 | MIN | 2.377 |
|                    | 60.40 | MAX | 2.378 |

Valve Seat Insert Bore Depth (7) and (8)

|                    | mm    |     | in    |
|--------------------|-------|-----|-------|
| Exhaust and Intake | 12.50 | MIN | 0.492 |
|                    | 12.62 | MAX | 0.497 |

If the valve guide seat insert inside diameter is **not** within specifications, use an oversize valve seat.

**Note : Oversize valve seat inserts with a pre-machined 20° seat angle are not available from Cummins Inc.**

Oversize valve seat inserts are available in the sizes listed in the table below. Machine the valve seat insert bore in the cylinder head to maintain a press fit within 0.064 to 0.114 mm [ 0.002 to 0.005 in ] between the valve seat insert and valve seat insert bore.

| Outside Diameter Oversize | Depth (Thickness) of Oversize |
|---------------------------|-------------------------------|
| 0.25 mm [ 0.010 in ]      | Standard                      |
| 0.51 mm [ 0.020 in ]      | 0.13 mm [ 0.005 in ]          |
| 0.76 mm [ 0.030 in ]      | 0.25 mm [ 0.010 in ]          |
| 1.02 mm [ 0.040 in ]      | 0.38 mm [ 0.015 in ]          |

The intake and exhaust valve face and seating surfaces **must** be cleaned and free of carbon deposits before the inspection is performed.

Use Scotch-Brite™ 7448 abrasive hand pad, Part Number 3823258, or equivalent, to clean the valve.

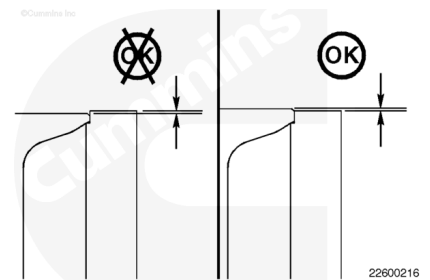
The valve head thickness gauge, Part Number 3164983, is used to check intake and exhaust valve head thickness. The lower gauging surface marked "INTAKE" is for the intake valve. The upper gauging surface marked "EXHAUST" is for the exhaust valve.

Place the valve into the valve head thickness gauge.

Measure the height of the valve.

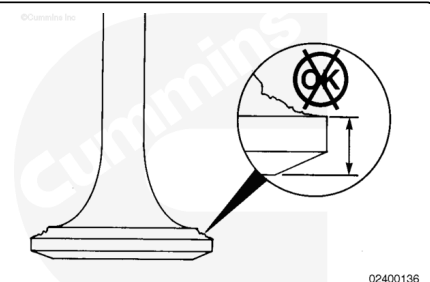
If the valve is flush or above the gauging surface, the valve can be used again.

If the valve is below the gauging surface, the valve can **not** be used again.



### ⚠ CAUTION ⚠

**This type of a check is not as accurate as the checking tool. It can result in valves that are not within minimum specification being reused. Valves that are not within minimum specification can fail, causing severe progressive damage to the power cylinder.**



If a valve checking tool is **not** available, place the valve on a flat surface and check the height of the outside diameter.

#### Minimum Valve Head Thickness (At the Outside Diameter)

|         | mm   |     | in    |
|---------|------|-----|-------|
| Exhaust | 3.00 | MIN | 0.120 |

|        |      |     |       |
|--------|------|-----|-------|
| Intake | 2.16 | MIN | 0.085 |
|--------|------|-----|-------|

If the valve is **not** within specification, the valve **must** be replaced.

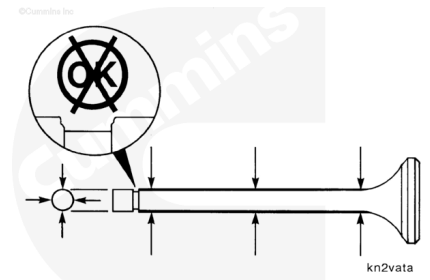
It is **not** necessary to remove any black or brown stain on the valve closest to the port, unless the buildup affects the movement in the guide. Use nothing more than coarse crocus cloth (1000 grit) to remove the stain.

The valve stems are plated with chrome. If there are scuffs or marks that can be felt with the fingernail, the valve **must** be replaced.

Check the collet groove area for wear. If the groove area is worn, the valve **must** be replaced.

Measure the valve stem outside diameter.

| Valve Stem Outside Diameter |     |       |
|-----------------------------|-----|-------|
| mm                          |     | in    |
| 12.535                      | MIN | 0.494 |
| 12.576                      | MAX | 0.495 |



Valve springs of three different lengths are used on the engine. The springs used with a valve rotator are called "short springs". Springs that are used when a rotator is **not** used are called "long springs".

There are two different lengths of "short springs". These are identified as red stripe (old) and white stripe (new) short springs.

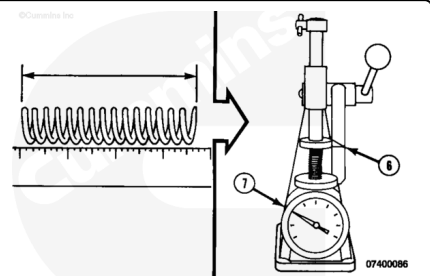
Cummins Inc. recommends using new white stripe valve springs when the cylinder head is reconditioned.

Do **not** reuse a bent or broken spring.

Do **not** reuse a spring with a notch worn in the first coil. The ends of the springs will wear into the first coil, creating a worn notch in the material.

If a worn notch can be seen or felt, do **not** reuse the valve spring.

Often a spring that has these notches will make a clicking sound. Hand compress the spring until the end of the first coil is completely collapsed to check for a click sound. A large





notch will cause a clicking sound.

Check both ends of the spring for the clicking sound.

Check the spring free length.

**Note : Natural gas engines use white stripe spring (with rotator).**

| Approximate Free Length          |                                     |                                          |
|----------------------------------|-------------------------------------|------------------------------------------|
| Long Spring<br>(without rotator) | Red Stripe Spring<br>(with rotator) | White Stripe<br>Spring (with<br>rotator) |
| 85 mm [ 3.35 in ]                | 65 mm [ 2.57 in ]                   | 69 mm [ 2.72 in ]                        |

Check the spring force at the indicated test height.

| Test Height                      |                                     |                                          |
|----------------------------------|-------------------------------------|------------------------------------------|
| Long Spring<br>(without rotator) | Red Stripe Spring<br>(with rotator) | White Stripe<br>Spring (with<br>rotator) |
| 50 mm [ 2.0 in ]                 | 39 mm [ 1.53 in ]                   | 38 mm [ 1.51 in ]                        |

| Valve Spring Force                             |      |     |     |
|------------------------------------------------|------|-----|-----|
|                                                | n    |     | lbf |
| Long<br>Spring<br>(without<br>rotator)         | 1053 | MIN | 237 |
|                                                | 1237 | MAX | 278 |
| Red Stripe<br>Spring<br>(with<br>rotator)      | 1183 | MIN | 266 |
|                                                | 1308 | MAX | 294 |
| White<br>Stripe<br>Spring<br>(with<br>rotator) | 1241 | MIN | 279 |
|                                                | 1383 | MAX | 311 |

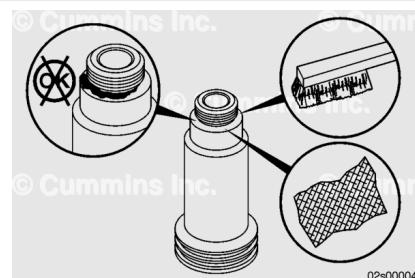
If the valve spring is **not** within specification, the valve spring **must** be replaced.

Inspect the cylinder head capscrew coating, especially under the capscrew head, and at the threads, for nicks, flakes, or missing spots.

If the capscrew coating is damaged or missing, the capscrew **must** be replaced.



Use the GTA38E G CMEICS G114 and GTA50E G CMEICS G115 Service Manual, Bulletin 5504185. Refer to Procedure 002-015 in Section 2 (/qs3/pubsys2/xml/en/procedures/445/445-002-015-tr.html) for cleaning and inspection of the spark plug adapter.



## Magnetic Crack Inspect

### ⚠ CAUTION ⚠

To reduce the possibility of engine damage, always demagnetize and clean the parts thoroughly after a magnetic particle inspection.

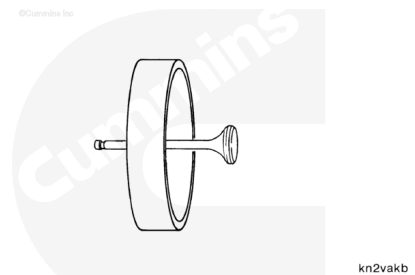
Use the magnetic particle residual method to check the valves for cracks.

Check the intake valves with the coil and head shot methods. Use the coil shot method first.

Check the exhaust valves with the coil shot method.

**Note :** Some natural gas fueled engines use an exhaust valve in all four locations.

Use a 305 mm [ 12 in ] minimum diameter coil.



### Coil Shot Amperage (Ampere Turns)

| Minimum                  | Maximum                  |
|--------------------------|--------------------------|
| 400-VDC or rectified VAC | 800-VDC or rectified VAC |

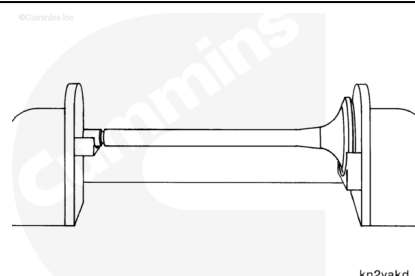
An ampere turn is a electrical current of one ampere flowing through the coil multiplied by the number of turns in the coil.

Test the valve.

A broad fuzzy pattern will appear at the welded joint on the exhaust valves. This is normal. If there is a distinct line in the broad fuzzy pattern, the valve **must** be replaced.

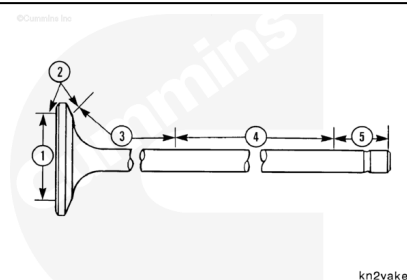
Test the intake valve with the head shot method.

| Head Shot Amperage (Ampere) |                          |
|-----------------------------|--------------------------|
| Minimum                     | Maximum                  |
| 500-VDC or rectified VAC    | 700-VDC or rectified VAC |



The acceptable criteria for intake and exhaust data is listed below:

1. Indications less than 38.1 mm [ 1.50 in ] in length are acceptable.
2. No indications.
3. **Only** longitudinal indications are acceptable.
4. **Only** longitudinal indications are acceptable.
5. No indications.

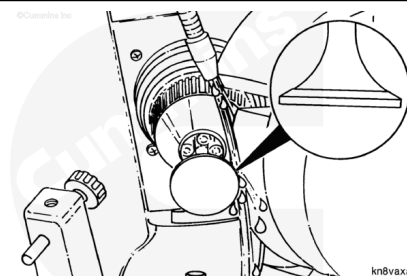


## Grind

### Valves

#### ⚠ CAUTION ⚠

Some natural gas fueled engines have valves and valve seats with a 20-degree angle. Some have intake valves and seats with 30-degree-30-second angle and exhaust valves and seats with a 20-degree angle. Verify the correct parts are used.



**Note :** Sursulf valves should not be lapped or ground. Sursulf valves are identifiable by the black coating and

an 'O' stamped in the center of the valve. Non-Sursulf valves can be lapped or ground, but this should not be necessary for new valves and seats.

Intake and exhaust valves **must** be ground to the same angle.

Use valve facing machine, Part Number 3376256, or equivalent, to grind the valve to a 30-degree angle, as shown.

Make sure **not** to remove too much material too quickly. **Only** remove the minimum amount of material, making sure the seating area of the valve is free from grooves.

## Machine

### Valve Seat

**Note :** Sursulf valve seat inserts should not be lapped or ground. Sursulf inserts are identifiable by the black coating of the insert and the part number. Non-Sursulf inserts can be lapped or ground, but this should not be necessary for new valves and seats.

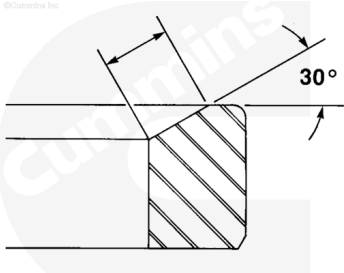
**Note :** Sursulf valve seat inserts are not used in the cylinder heads for natural gas fueled engines.

Cummins Inc. recommends the use of machines that will cut the valve seat insert. Grinding the valve seat with stones that are too coarse can cause an unacceptable surface finish that can lead to early valve and/or seat wear. If the valve protrusion is too great and a machine to cut the valve seat is **not** available, Cummins Inc. recommends to remove the seat then machine the bore in the cylinder head to a greater depth, if possible.

Use a valve guide arbor set, Part Number 3375946, or equivalent, with valve seat grinding machining ST-685-A (110-VAC) or ST-685-C (220-VAC).

Machine the valve seat insert to the correct angle for the cylinder head being serviced.

Measure the valve seat insert width for a diesel cylinder head.



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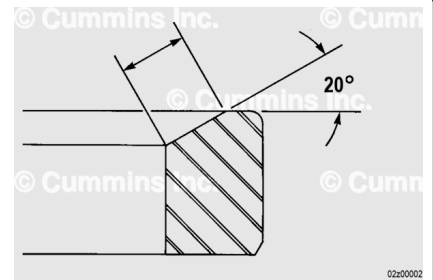
Valve Seat Insert Width (Diesel)

|        | mm   |     | in    |
|--------|------|-----|-------|
| Intake | 3.05 | MIN | 0.120 |
|        | 3.55 | MAX | 0.140 |

|         |      |     |       |
|---------|------|-----|-------|
| Exhaust | 1.52 | MIN | 0.060 |
|         | 2.54 | MAX | 0.100 |

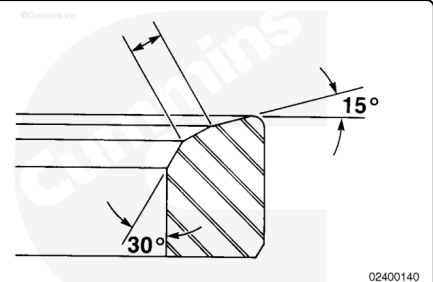
Measure the valve seat insert width for a natural gas head.

| Valve Seat Width (Natural Gas) |      |     |       |
|--------------------------------|------|-----|-------|
|                                | mm   |     | in    |
| Intake                         | 5.33 | MIN | 0.210 |
|                                | 5.71 | MAX | 0.225 |
| Exhaust                        | 5.33 | MIN | 0.210 |
|                                | 5.71 | MAX | 0.225 |



If the width of the valve seat is **not** within specifications, remove the surface material on the inside diameter and outside diameter of the valve seating area to decrease the width of the valve seat.

If the valve insert seat specifications are **not** obtained by grinding, the valve insert **must** be replaced.



Do **not** position the eccentricimeter too far down the arbor. The meter needle **must** complete less than one to one and a half revolutions before touching the valve seat insert. If the meter will **not** touch the insert at a minimum revolution, change the arbor to a smaller diameter. The meter will have to be adjusted for each seat and arbor combination.

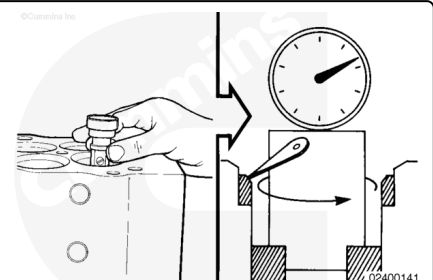
Use an eccentricimeter, Part Number ST-685-4, or equivalent, and the correct arbor from valve seat grinding machine kit, Part Number ST-685-A, or equivalent, or Part Number ST-685-C, or equivalent.

**Note :** When using the eccentricimeter, the pin that contacts the valve seat **must** always remain on the sealing surface of the seat, or incorrect measurements will be recorded.

Measure the valve seat to the valve guide concentricity.

The seat and guide **must** be concentric within 0.05 mm [ 0.002 in ].

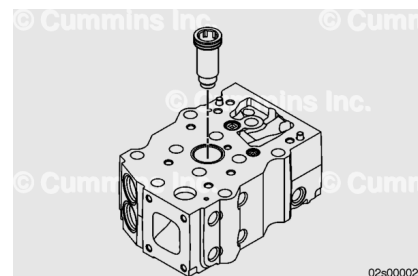
If the concentricity is **not** within specification, machine the valve seat.



# Assemble

## with Mechanically Actuated Injector

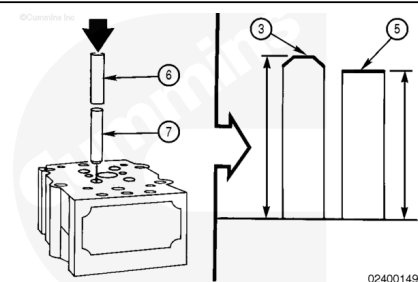
Install the spark plug adapter. Use the GTA38E G CMEICS G114 and GTA50E G CMEICS G115 Service Manual, Bulletin 5504185. Refer to Procedure 002-015 in Section 2 (/qs3/pubsys2/xml/en/procedures/445/445-002-015-tr.html).



Install the valve guides in all four locations in the cylinder head.

Use the appropriate valve guide driver (6) and an arbor press to install the valve guide (7) to the specified height.

- Taper-top valve guide (3), use valve guide driver, Part Number 3376779, or equivalent.
- Flat-top valve guide (5), use valve guide driver, Part Number 3376149, or equivalent.



Valve Guide Height (Installed)

|                  | mm     |     | in    |
|------------------|--------|-----|-------|
| Taper-Top<br>(3) | 33.665 | MIN | 1.325 |
|                  | 34.163 | MAX | 1.345 |
| Flat-Top<br>(5)  | 29.210 | MIN | 1.150 |
|                  | 29.178 | MAX | 1.170 |

**Note :** This specification applies to the valve guides on a natural gas fueled engine that has taper-top guides in the exhaust location and flat-top guides in the intake position.

Valve Guide Height (Installed) Natural gas engines with four flat-top guides

|              | mm    |     | in    |
|--------------|-------|-----|-------|
| Flat Top (5) | 27.61 | MIN | 1.087 |
|              | 28.37 | MAX | 1.117 |

Valve Guide Height (Installed) Natural gas engines  
with 30.5 mm [ 1.202 in ] long spark plug adapter.

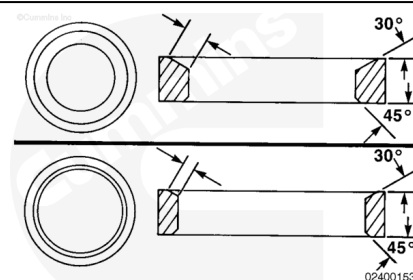
|                        | mm    |     | in    |
|------------------------|-------|-----|-------|
| Flat-Top<br>(Intake)   | 33.52 | MIN | 1.320 |
|                        | 34.28 | MAX | 1.349 |
| Taper-Top<br>(Exhaust) | 33.28 | MIN | 1.310 |
|                        | 34.04 | MAX | 1.340 |

Identify the correct valve seat insert for diesel cylinder heads.

- The top illustration is the intake port valve seat insert with a wide valve seating area.
- The bottom illustration is the exhaust port valve seat insert.

Both the intake and exhaust port valve seat inserts **must** be installed with the 30-degree angle positioned as illustrated.

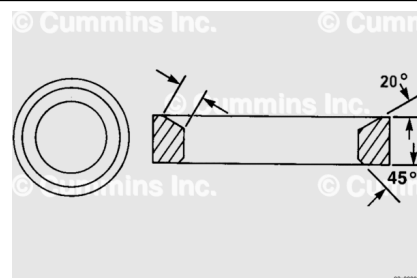
Exhaust valve seat inserts are color coded blue to identify the 30-degree angle side of the seat insert.



Identify the correct valve seat for natural gas cylinder heads.

Both the intake and exhaust valve seat inserts **must** be installed with the 20-degree or 30-degree angle positioned as illustrated.

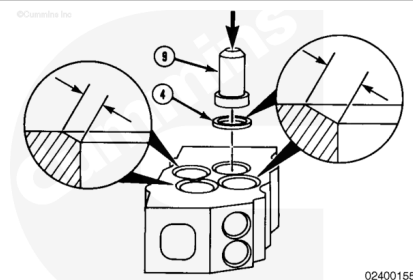
The part number is etched on the outside diameter of the insert. Verify the correct part number is used.



### ⚠ CAUTION ⚠

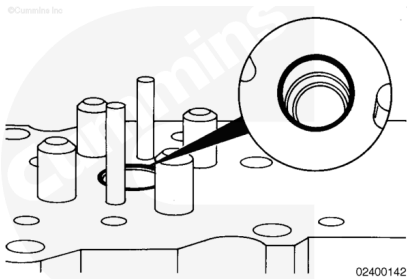
**Do not use a hammer or mallet to install the valve seat inserts. The inserts can be cracked leading to a valve seat failure.**

Install the valve seat insert (4) with a mandrel (9) and an arbor press.



Check the injector bore for cracks or other damage.

If cracks or other damage is found, the cylinder head **must** be repaired or replaced.

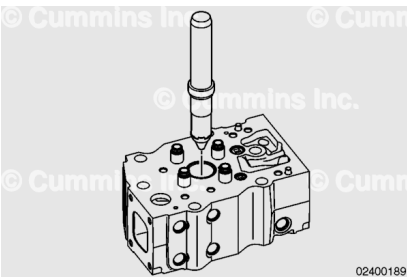


**⚠ CAUTION ⚠**

**Support the cylinder head to prevent damage to the injector tip that protrudes from the combustion face.**

Install the injector protrusion gauge, Part Number 4919196, into the cylinder head. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html>)

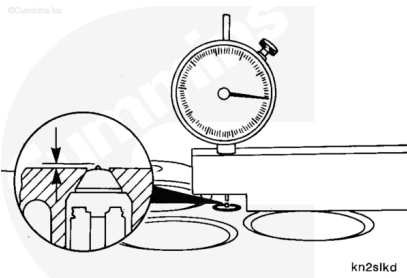
**Note :** There is no requirement to measure spark plug protrusion on a natural gas cylinder head.



Turn the cylinder head over.

Measure the injector protrusion with a depth gauge, Part Number 3164438.

| Injector Protrusion |     |       |
|---------------------|-----|-------|
| mm                  |     | in    |
| 2.29                | MIN | 0.090 |
| 2.79                | MAX | 0.110 |



If the protrusion is **not** within specifications, use an oversize sealing ring.

The sealing ring sizes and the amount of injector protrusion associated with the use of each ring is listed in the table below.

| Injector Sealing Rings                   |                            |
|------------------------------------------|----------------------------|
| Wall Thickness                           | Injector Protrusion Change |
| 0.343 to 0.419 mm [ 0.014 to 0.017 in ]* | 0.00 mm [ 0.000 in ]       |
| 0.470 to 0.546 mm [ 0.019 to 0.022 in ]  | 0.25 mm [ 0.010 in ]       |
| 0.597 to 0.673 mm [ 0.024 to 0.027 in ]  | 0.51 mm [ 0.020 in ]       |



## Injector Sealing Rings

### Wall Thickness

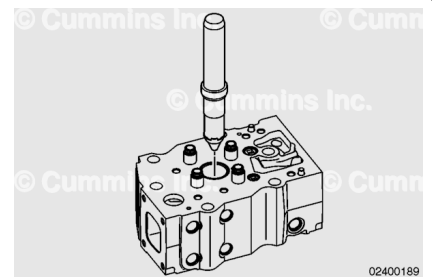
0.724 to 0.800 mm [ 0.029 to 0.032 in ]

### Injector Protrusion Change

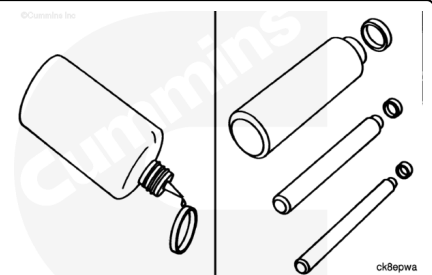
0.76 mm [ 0.030 in ]

\* The injector sealing ring with a wall thickness of 0.343 to 0.419 mm [ 0.014 to 0.017 in ] is the standard sealing ring.

Remove the injector protrusion gauge, Part Number 4919196, or equivalent, from the cylinder head.



Apply Loctite™ 609 sealant, Part Number 3823718, or equivalent, to the cup plugs.

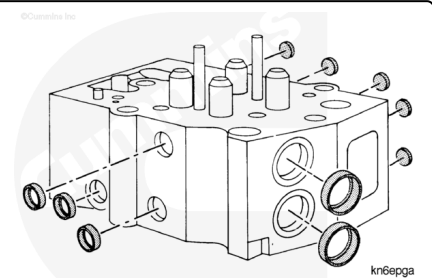


Use the correct expansion plug driver to make sure the expansion plugs are installed to the correct depth in the cylinder head. Drive the plug until the shoulder of the driver contacts the cylinder head.

- Part Number 3376813, expansion plug driver
- Part Number 3376814, expansion plug driver.

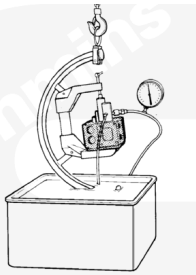
A cup plug driver handle, Part Number 3164085, or equivalent, **must** be used with expansion plug driver, Part Number 3376813, or equivalent, and expansion plug driver, Part Number 3376814, or equivalent.

Install the 10 new cup plugs.



Use a hydrostatic tester to check for evidence of leaks around the cup plugs.

Reference the Pressure Test section in this procedure.

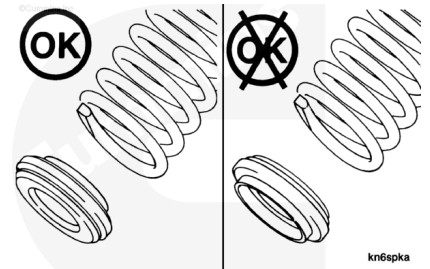


kn6towa

When turning the rotator by hand, it can turn roughly, be difficult to turn, or **not** turn at all. This is normal until the rotator is installed. After installation, it will rotate freely.

The rotator **must** be soaked in clean engine oil for at least 15 minutes prior to installation.

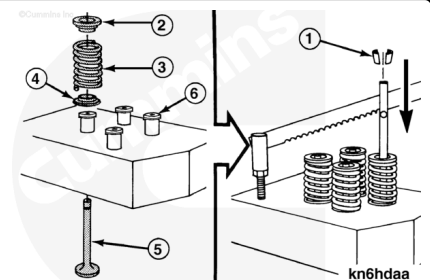
The rotator **must** be installed over the valve guide with the spring pilot flange face up, as illustrated.



kn6spka

### **⚠ WARNING ⚠**

**The valve springs are under compression. Use care when using the valve spring compressor. Personal injury can result if the tool slips from your hands.**



kn6hdaa

Use a valve spring compressor, Part Number 3375960, or equivalent, valve spring stand, Part Number ST-1022, or equivalent, and compressor plate, Part Number 3375043, or equivalent, to compress all four springs at the same time.

Install the parts.

1. Valve collet
2. Valve rotator
3. Valve spring
4. Valve spring retainer
5. Valve
6. Valve stem guide

Valve stem seal (if required) **not** shown.

**Note :** Service tool, Part Number 4918258, or equivalent, is used to install both the valve guide to the proper height and install the valve stem seals as well.

Install the valve stem seal. Use valve guide installation tool, Part Number 4918258, or equivalent.

Thoroughly lubricate the valve guide inside diameter with 140 weight gear oil before installing the valves.

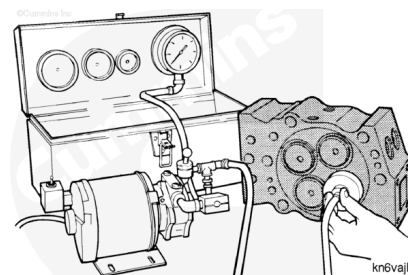
Use a valve vacuum tester, Part Number 3824277, or equivalent.

If the valve to the valve seat contact pattern appears to be correct and the valve still does **not** pass the vacuum test, perform the following check.

Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.

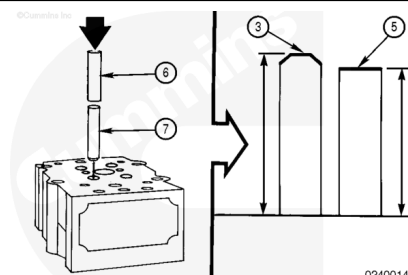
If the leakage is between the insert and the valve, repair the valve seat.

If the leakage is between the insert and the head, the insert **must** be replaced.



## with Electronically Actuated Injector

Use the valve guide installation tool (6), Part Number 4918258, or equivalent, and an arbor press to install the valve guide (7) to the specified height.



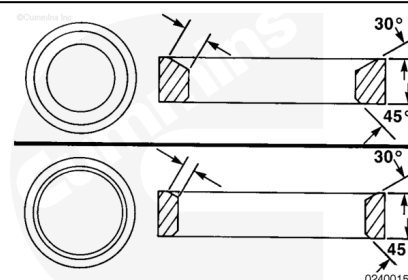
| Valve Guide Height (Installed) |     |       |
|--------------------------------|-----|-------|
| mm                             |     | in    |
| 27.737                         | MIN | 1.092 |
| 28.245                         | MAX | 1.112 |

Identify the correct valve seat insert.

- The top illustration is the intake port valve seat insert with a wide valve seating area.
- The bottom illustration is the exhaust port valve seat insert.

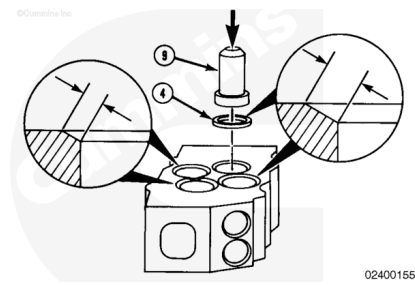
Both the intake and exhaust port valve seat inserts **must** be installed with the 30-degree angle positioned as illustrated, with the 45-degree angle to the cylinder head port.

Exhaust valve seat inserts are color coded blue to identify the 30-degree angle side of the seat insert.



**⚠ CAUTION ⚠**

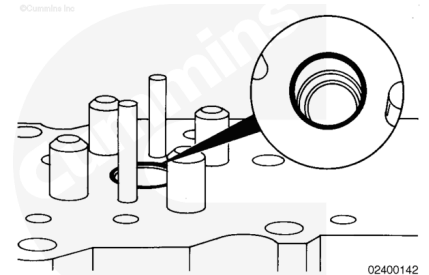
**Do not use a hammer or mallet to install the valve seat inserts. The inserts can be cracked leading to the seat failure.**



Install the valve seat insert (4) with a mandrel (9) and an arbor press.

Check the injector bore for cracks or other damage.

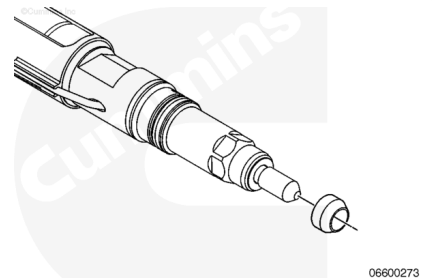
If cracks or other damage is found, the cylinder head **must** be repaired or replaced.



Install a sealing ring on a dummy injector.

Do **not** install the injector o-rings.

**Note :** There is no injector protrusion check or protrusion specifications on electronically actuated injectors, as injector protrusion can **not** be adjusted.



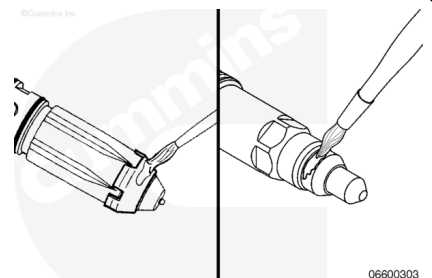
### **⚠ CAUTION ⚠**

**Support the cylinder head to prevent damage to the injector tip that protrudes from the combustion face.**

Apply a thin coating of Prussian Blue into the sealing ring.

Install the injector into the cylinder head. Refer to Procedure 006-026 (Injector) in Section 6.

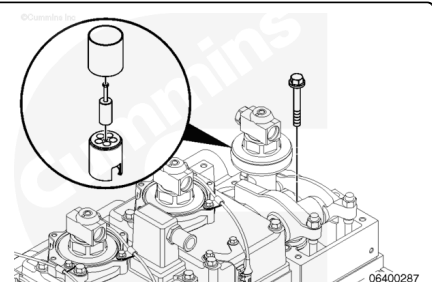
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Turn the cylinder head over to the original position.

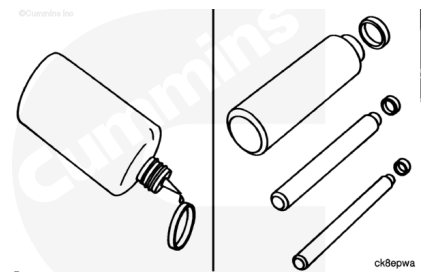
Remove the injector hold-down clamp and the injector.

Check the contact pattern in the injector bore. A blue band, a minimum of 1.52 mm [ 0.060 in ] wide and 360 degrees full circumference, **must** be visible. If the pattern does **not** meet specifications, clean the bore of the injector with a tapered brush and check the sealing pattern.



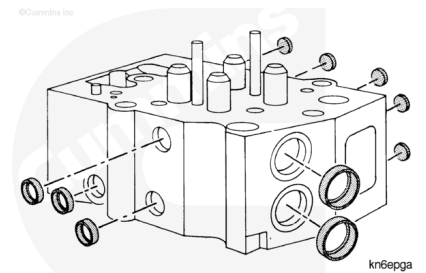
The cylinder head **must** be replaced if the injector pattern is **not** within specifications.

Apply Loctite™ 609 sealant, Part Number 3823718, or equivalent, to the cup plugs.



Cup plug driver handle, Part Number 3164085, or equivalent, is used with expansion plug drivers, Part Number 3376813, or equivalent, and Part Number 3376814, or equivalent.

Use the appropriate expansion plug driver and handle combination to drive the cup plug in until the shoulder of the driver contacts the cylinder head.

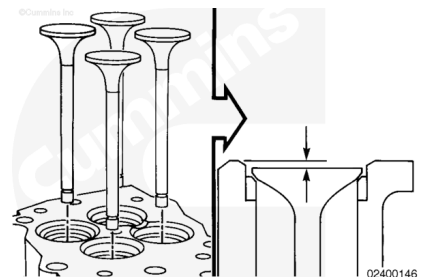


Install the valve into the valve guide.

Measure the depth of the valve with a depth gauge, Part Number 3164438, or equivalent.

The valve head **must** be even with, or **not** more than 0.51 mm [ 0.020 in ] below, the surface of the cylinder head.

If the valve depth is **not** within specifications, the valve or the valve seat **must** be replaced.



**Note :** Sursurfl valves should **not** be lapped or ground. Sursurfl valves are identifiable by the black coating and an 'O' stamped in the center of the valve.

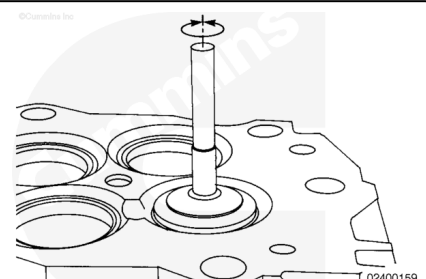
**Note :** Non-Sursurfl valves can be lapped or ground, but this should **not** be necessary for new valves and seats.

Apply a thin, even coating of fine lapping compound, Part Number 3375805, or equivalent, onto the valve face.

Provide pressure in the center of the valve with a power or hand suction lapping tool.

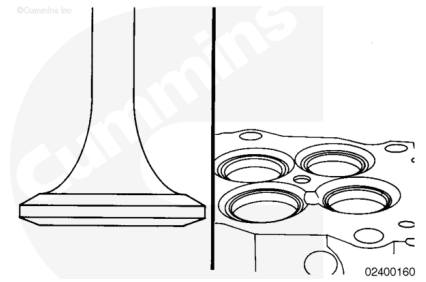
Turn the valve backward and forward.

Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.



**⚠ CAUTION ⚠**

**Lapping compound is an abrasive material. Failure will result if the cylinder head, the valves, and the valve seats are not cleaned thoroughly.**



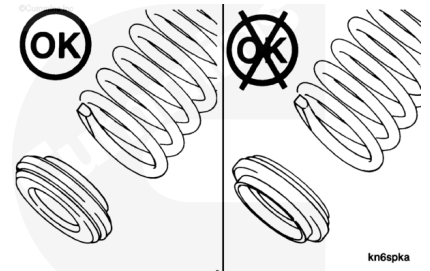
Remove the valve.

Clean the lapping compound from the valve, valve seat insert, and cylinder head.

When turning the rotator by hand, it can turn roughly, be difficult to turn, or **not** turn at all. This is normal until the rotator is installed; after installation it will rotate freely.

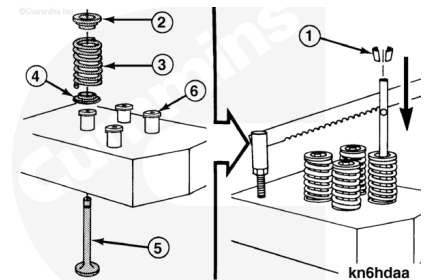
The rotator **must** be soaked in clean engine oil for at least 15 minutes prior to installation.

The rotator **must** be installed over the valve guide with the spring pilot flange face up, as illustrated.



**⚠ WARNING ⚠**

**The valve springs are under compression. Use care when using the valve spring compression tool. Personal injury can result if the tool slips out of the hands.**



Use one of the below listed valve spring compressors to install the valve springs:

- Valve spring compressor, Part Number 3163606, or equivalent
- Valve spring compressor stand, Part Number ST-1022, or equivalent
- Valve spring compressor plate, Part Number 3163177, can be used with valve spring compressor, Part Number 3163606, or equivalent, and valve spring compressor stand, Part Number ST-1022, to remove four springs at the same time
- Air operated valve spring compressor, Part Number 3375960.

Thoroughly lubricate the valve guide inside diameter with 140 weight gear oil.

Install the parts:

1. Valve collet

2. Valve rotator/or guide
3. Valve spring
4. Valve spring retainer
5. Valve
6. Valve stem guide

**Note :** Service tool, Part Number 4918258, or equivalent, is used to install both the valve guide to the proper height and install the valve stem seals as well.

Install the valve stem seal. Use valve guide installation tool, Part Number 4918258, or equivalent.

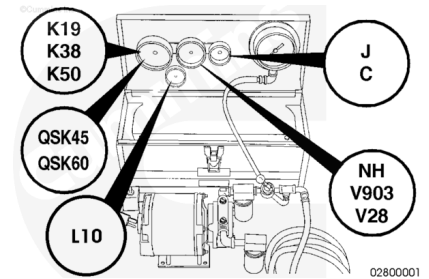
## Vacuum Test

The valve vacuum tester, Part Number 3824277 (115-VAC, 50/60 hz), or 3824278 (220-VAC, 50/60 hz), can be used to test all Cummins® engine models.

Before using the tester, test the leakage shutoff valve. If the valve is dirty or worn, it will produce a false leakage measurement on the gauge.

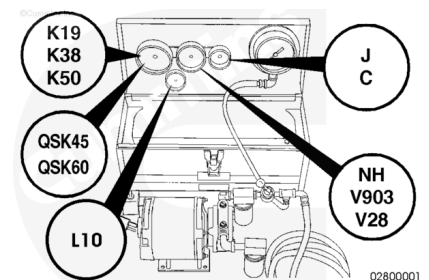
Check the valve:

- Open the shutoff valve
- Turn on the vacuum pump
- Place the cup against a smooth surface
- Close the shutoff valve
- Turn off the vacuum pump
- Wait approximately 10 seconds
- The gauge **must not** drop more than 7 kPa [ 2 in-Hg ].

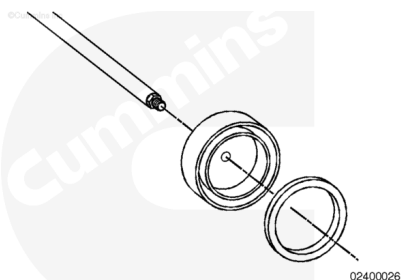


The valve and the valve seats **must** be clean and dry.

Choose the correct cup and the correct seal from the service tool kit for the engine model that is to be tested.

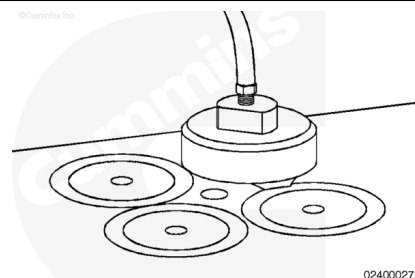


Install the seal and the cup to the vacuum line (hose).



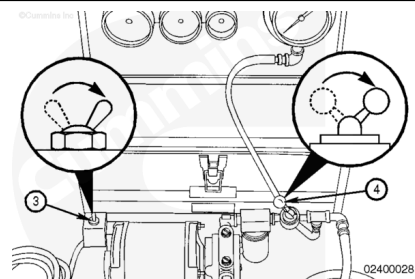
Cover the valve with the cup and the seal. The seal **must** have a tight contact on the cylinder head around the valve.

To check the exhaust valves, the seal **must** completely fill the milled area between the exhaust valves.



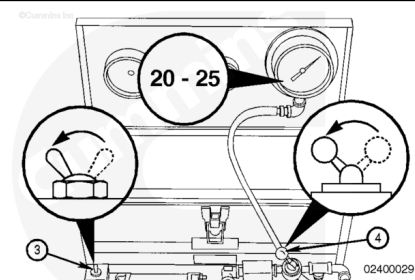
Move the toggle switch (3) to the ON position.

Turn the vacuum control valve (4) to the OPEN position.



When the gauge indicates between 34 to 85 kPa [10 to 25 in Hg], turn the vacuum control valve (4) to the CLOSED or OFF position.

Turn the toggle switch (3) to the OFF position.

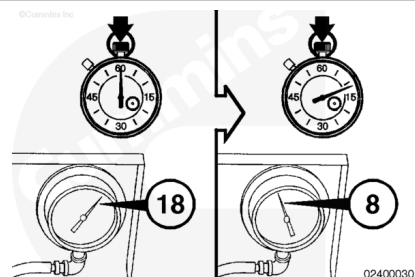


### ⚠ CAUTION ⚠

**The cylinder head must be disassembled and cleaned after any grinding or cutting procedures to reduce the possibility of engine damage.**

Use a stopwatch. As the vacuum gauge needle moves **counterclockwise**, start timing when the needle on the gauge points to 61 kPa [ 18 in-Hg ].

Stop timing when the needle on the gauge points to 34 kPa [ 10 in-Hg ].





The elapsed time for the gauge to move between the specified readings **must** be 3 seconds or more.

If the elapsed time is less than 3 seconds, perform the following checks:

- Repeat the test to be certain the equipment is functioning properly.
- Use a mallet to lightly hit the valve stem to be certain the valve is sealed. Repeat the test.
- Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.
- If the leakage is between the valve insert and the head, the insert **must** be replaced.

If the leakage is between the valve and the valve insert seat, one of the following procedures **must** be performed:

- Lap valve to insert seat
- Grind the valves
- Grind the valve insert seat.

## Install

### **⚠ WARNING ⚠**

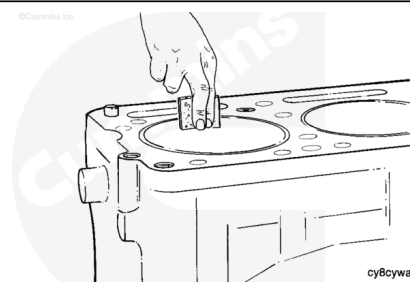
**Some solvents are flammable and toxic. Read the manufacturer's instructions before using.**

### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

### **⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

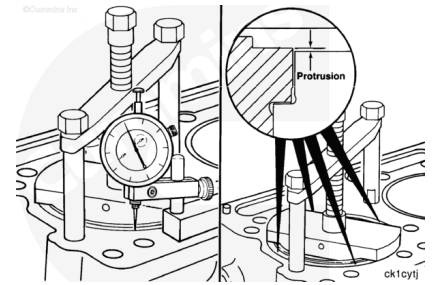


Clean the top of the cylinder block and the cylinder liners.

Clean the threaded holes in the cylinder block with contact cleaner, Part Number 3824510, or equivalent.

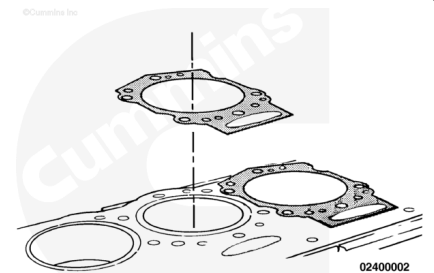
Dry with compressed air.

Measure the liner protrusion. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html>)



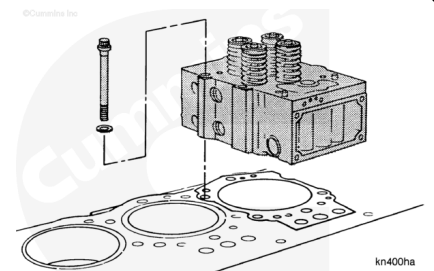
The word TOP, stamped on top of the head gasket, **must** be visible after the gasket is installed.

Install the gasket.



**⚠ WARNING ⚠**

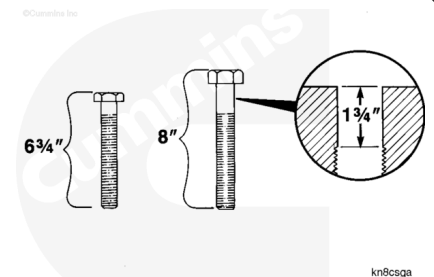
**This component or assembly weighs greater than 23 kg [ 50 lb ]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



Install the cylinder head. It **must** fit easily over the dowel pins

**⚠ CAUTION ⚠**

**Do not combine black and silver capscrews in any one cylinder head. The torque requirements are different.**



**⚠ CAUTION ⚠**

**Install capscrews of the correct length or damage to the cylinder block will result.**

**Note :** Two different capscrew coatings have been used on the 170 mm [ 6.75 in ] capscrews used with some K38 cylinder heads. The current capscrews are

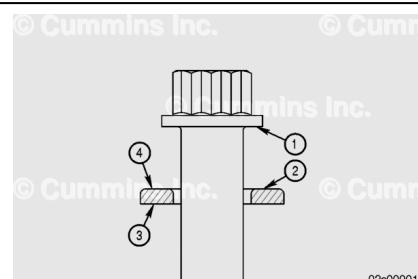
black. The previous capscrews were silver, and are no longer available.

**Note :** Two different lengths of cylinder head capscrews have been used on the K38 engines. Engines with an Engine Serial Number (ESN) greater than 33101470 and all service blocks have capscrews that are 203 mm [ 8 in ] in length. Engines with an ESN less than 33101470, have capscrews that are 170 mm [ 6.75 in ] in length. The capscrews that are 203 mm [ 8 in ] in length (black) are used in the K50 engines.

If the capscrew length is questionable, check the distance from the top of the cylinder block to the first thread in the hole for the cylinder head capscrew. If the distance is approximately 44 mm [ 1.75 in ], the 203 mm [ 8 in ] capscrews must be installed.

**⚠ CAUTION ⚠**

**Allow the excess oil to drip from the capscrews before installing into the cylinder block or damage to the cylinder block may result.**



Apply clean engine oil to the threads of the cylinder head capscrews.

Apply clean engine oil to both sides of the washer (2)

Install the capscrews.

Use the following steps to tighten the capscrews in the sequence shown.

**Torque Value:**

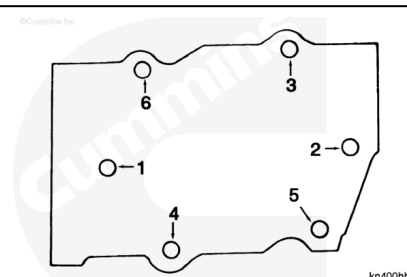
Black - 170 mm [ 6.75 in ] and 203 mm [ 8 in ]

1. 271 n•m [ 200 ft-lb ]
2. 407 n•m [ 300 ft-lb ]
3. 556 n•m [ 410 ft-lb ]

**Torque Value:**

Silver (170 mm [ 6.75 in ] **ONLY**)

1. 65 n•m [ 48 ft-lb ]
2. 160 n•m [ 118 ft-lb ]
3. 250 n•m [ 185 ft-lb ]
4. 345 n•m [ 255 ft-lb ]

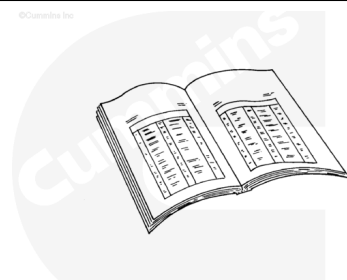


# Finishing Steps

## with Mechanically Actuated Injector

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



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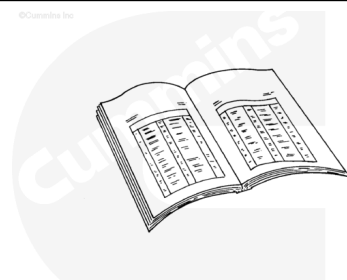
- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html>)
- Install the rocker lever housings, push rods, and crossheads. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html>)
- Install the high-pressure fuel lines to the injectors. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html>)
- Install the push rods or tubes. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html>)
- Adjust the valves. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Install the gear cover clamping plate. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html>)
- Install the coolant thermostat housing support. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-015-tr.html>)

- Install the fuel tubes. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-024-tr.html)
- Install the fuel manifolds. Refer to Procedure 006-022 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-022.html)
- Install the aftercooler. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html)
- Fill the cooling system. Refer to Procedure Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect air starter supply line, if equipped. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine to 70°C [ 160°F ] minimum coolant temperature and check for leaks.

## with Electronically Actuated Injector

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



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- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-026-tr.html)
- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-013-tr.html)

- Install the push rods or tubes. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/28/28-004-014-tr.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html>)
- Adjust the overhead. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Install the fuel drain lines. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-013.html>)
- Install the aftercooler. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-002-tr.html>)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html>)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-033-tr.html>)
- Connect the batteries. Refer to the OEM service manual.
- Connect the air starter supply line, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Operate the engine to 70°C [ 160°F ] minimum coolant temperature and check for leaks.